

MATHEWS V MANOJ

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PROFESSIONAL SUMMARY

Third-year Electronics and Communication Engineering student at Muthoot Institute of Technology and Science (**CGPA: 9.25/10.0**) with hands-on experience in embedded systems, FPGA design, and IoT solutions. **e-Yantra Regional Finalist** with expertise in VHDL/Verilog, Arduino, ESP32, and ZedBoard platforms. **Founder of Focuno**, an AI-powered productivity platform. Aspiring **GATE** candidate with strong analytical skills and passion for defense technology entrepreneurship.

EDUCATION

Bachelor of Technology in Electronics and Communication Engineering

Muthoot Institute of Technology and Science, APJ Abdul Kalam Technological University (KTU)

Expected Graduation: **2028** | CGPA: **9.25/10.0**

Location: Kerala, India

EXPERIENCE

Founder and Developer

2024 - Present

Focuno

- Founded and leading development of AI-powered focus coaching platform to help users maintain deep concentration through intelligent nudges and attention management strategies
- Managing end-to-end product development including user research, feature design, and technical implementation

INTERNSHIPS

Antenna Research Intern

National Institute of Technology, Calicut (NITC)

- Researched and designed a **broadband blade monopole antenna** spanning **20 MHz - 2500 MHz** for electronic warfare applications, addressing wideband impedance matching and miniaturization challenges

5G Hybrid Internship

Indian Institute of Space Science and Technology (IIST), Trivandrum

TECHNICAL SKILLS

Programming Languages: Python, C, Embedded C, Verilog, VHDL

Development Tools: Arduino IDE, Xilinx Vivado, NumPy, Pandas, CST Studio Suite, KiCad, MATLAB

Hardware Platforms: ZedBoard (Zynq-7000 FPGA), Arduino, ESP32, Raspberry Pi

Core Competencies: Embedded Systems Design, FPGA Programming, Digital Circuit Design, IoT Development, Machine Learning Fundamentals, Real-Time Systems, Hardware-Software Integration

PROJECTS

Broadband Blade Monopole Antenna for Airborne Electronic Warfare Systems

Technologies: CST Microwave Studio, VBA, Antenna Design, RF/Microwave Engineering

- Designed a broadband blade monopole antenna spanning **20 MHz - 2500 MHz** (125:1 bandwidth ratio) for airborne electronic warfare systems
- Simulated and validated antenna performance in CST Microwave Studio, evaluating FEM, Time Domain, and Integral Equation solvers to balance accuracy against computational cost
- Developed and debugged VBA macros for automated parametric sweeps, including implementation of Wu-King resistive loading to extend operating bandwidth
- Extended the antenna's low-frequency edge from 127 MHz to **80 MHz** within 16 GB RAM constraints, and benchmarked design performance against Chu and Bode-Fano gain-bandwidth limits

RestorationNet — Deep Learning for Image Restoration

Technologies: PyTorch, Python

- Designed and trained a **0.96M-parameter** residual CNN performing joint denoising and 2x super-resolution on degraded semiconductor inspection images (128x128 to 256x256), achieving **27.99 dB PSNR, 0.769 SSIM, 0.303 LPIPS**
- Selected the final architecture through a controlled ablation against channel-attention (RCAB) and RRDB variants, demonstrating that a compact residual network outperformed heavier models on this dataset
- Identified a domain-specific data insight - that out-of-range noisy pixel values carry signal and must not be clipped - which directly improved restoration accuracy
- Benchmarked end-to-end inference at **18.3 ms/image (54.6 images/sec)** on an RTX 5050, with warm-up discarded and CUDA-synchronised timing over 400 samples
- Delivered a fully reproducible pipeline - standalone inference script, committed checkpoints, pinned dependencies - runnable offline with no manual configuration

Hazard and Fault Detection System with Real-Time Display

Technologies: Verilog HDL, ZedBoard (Zynq-7000 FPGA), Xilinx Vivado

- Designed Verilog HDL-based system for real-time hardware hazard and fault detection on ZedBoard FPGA platform
- Integrated real-time status monitoring using onboard peripherals for immediate fault visualization
- Verified design through comprehensive simulation testbenches and validated timing and fault-detection logic on hardware

Spoilage Detection and Prevention System

Technologies: ESP32, Sensors, Embedded C

- Engineered IoT-based monitoring system using ESP32 microcontroller and environmental sensors for real-time spoilage detection
- Implemented threshold-based alert logic for continuous parameter tracking and early prevention of food spoilage

Obstacle Avoidance Robot

Technologies: Arduino, Embedded C, Ultrasonic Sensors

- Built autonomous robot with ultrasonic sensor array for real-time obstacle detection and intelligent motor control system

Machine Health Prediction using Markov Chain

Technologies: Python, NumPy

- Developed probabilistic Markov Chain model to predict system state transitions for predictive maintenance applications

CERTIFICATIONS

- NPTEL - Robotics (Elite Certification)
- NPTEL - Microsensors and Nanosensors (Elite Certification)
- NIELIT - Python Programming
- NIELIT - Embedded Systems

ACHIEVEMENTS

- **e-Yantra Innovation Challenge** - Regional Finalist (2025-26)
- **National Road Safety Hackathon** (2025) - Advanced to Stage 2 of 4
- **MYOSA Development Stage** - Selected

POSITIONS OF RESPONSIBILITY

- **Mathematics Club Coordinator**, Muthoot Institute of Technology and Science